

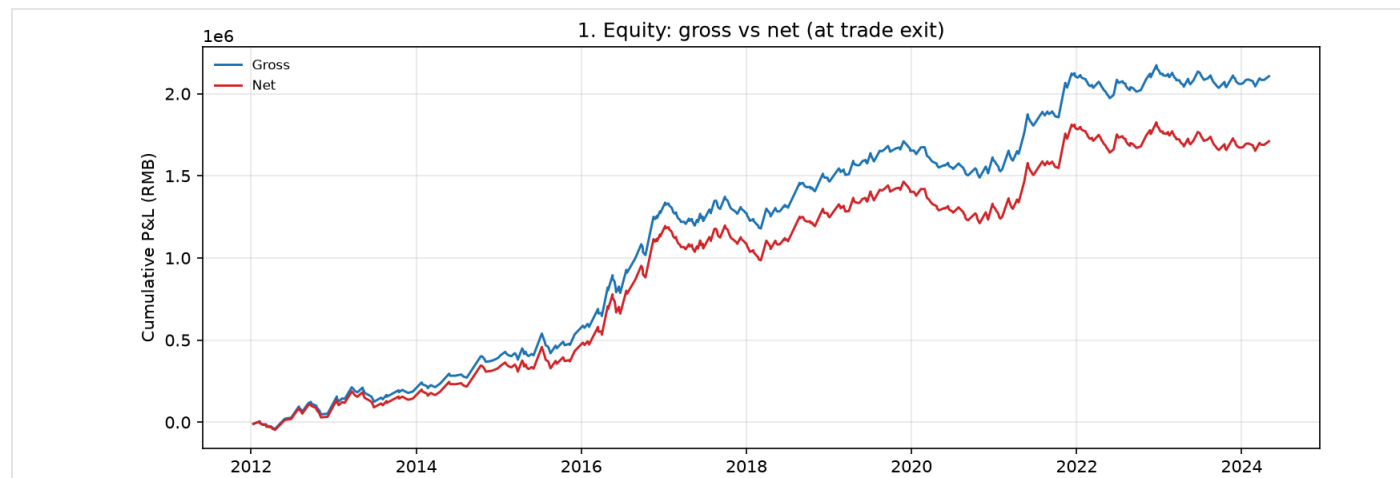
Systematic Strategy Research

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Two systematic trading research projects, each taken through to a statistically validated verdict: first the development of a day-trading strategy alongside an investigation into overfitting and backtest-validation methodology, then a cross-sectional factor research and generation project spanning hundreds of feature hypotheses.

Day-Trading Strategy — SHFE Rebar Futures

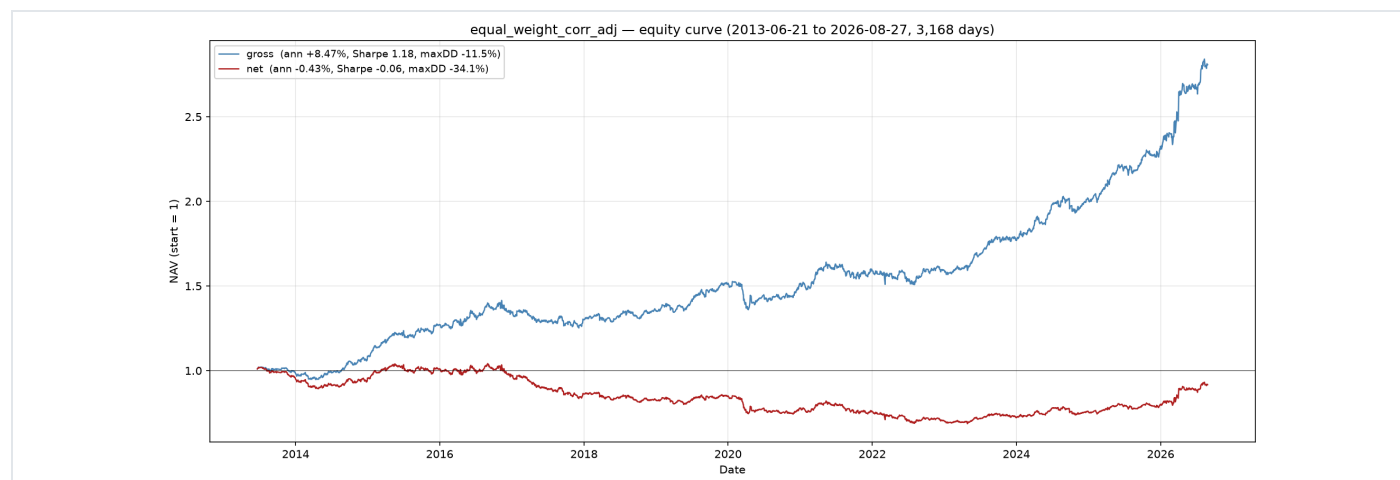
- Built an EMA-channel breakout strategy and a bar-level backtest engine, exploring the channel, signal, and entry-mechanics parameter space to an in-sample Sharpe of **1.03** over 12 years (+171%).
- Tested the result against a battery of checks: Monte Carlo resampling (bootstrap, shuffle, parametric), a **deflated Sharpe ratio**, and a walk-forward across rolling and expanding training windows.
- Confirmed on a true out-of-sample window (2024–2026): **−11.4%**, validating the overfitting diagnosis.



In-sample equity, gross vs net (2012–2024, +171% net).

Cross-Sectional Intraday Factor Research — Chinese Commodities Futures

- Generated and tested **860 candidate features** (964 variants) spanning minute-bar price action, order flow, open interest, and tick-level microstructure — distinct hypotheses about what predicts next-day relative performance across **58 products** in the Chinese commodities futures complex.
- **43 variants survived** a Romano–Wolf stepwise multiple-testing filter and a year-on-year stability screen, including features on trend quality, VWAP deviation, and timing, and were combined into a long-short book with a correlation-adjusted weighting scheme that down-weights redundant, highly correlated variants.
- The book carried a strong **gross Sharpe of ~1.2** (t up to 4.4), reduced to roughly zero net by transaction costs at the turnover required — a result useful as a low-correlation return stream that can smooth drawdowns when blended into an existing strategy, and a foundation to build on: more features and a stricter turnover/cost filter are the natural next steps toward net edge.



Cross-sectional book equity, gross vs net (2013–2026).